

Customer Shopping Behaviour Analysis

1. Project Overview: This project analyses customer shopping behaviour using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behaviour to guide strategic business decisions.

2. Dataset Summary:

2.1. Rows: 3,900

2.2. Columns: 18

2.3. Key Features: -

2.3.1. Customer demographics (Age, Gender, Location, Subscription Status)

2.3.2. Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, colour)

2.3.3. Shopping behaviour (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)

2.3.4. Missing Data: 37 values in Review Rating columns.

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- Data Loading: Imported the dataset using pandas.
- Initial Exploration: Used `df.info()` to check structure and `.describe()` for summary statistics.

[5]: `df.info()`

```
<class 'pandas.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                          3900 non-null   int64
1   Age                                  3900 non-null   int64
2   Gender                              3900 non-null   str
3   Item Purchased                       3900 non-null   str
4   Category                            3900 non-null   str
5   Purchase Amount (USD)                3900 non-null   int64
6   Location                             3900 non-null   str
7   Size                                 3900 non-null   str
8   Color                                3900 non-null   str
9   Season                               3900 non-null   str
10  Review Rating                         3863 non-null   float64
11  Subscription Status                   3900 non-null   str
12  Shipping Type                         3900 non-null   str
13  Discount Applied                     3900 non-null   str
14  Promo Code Used                       3900 non-null   str
15  Previous Purchases                    3900 non-null   int64
16  Payment Method                       3900 non-null   str
17  Frequency of Purchases                3900 non-null   str
dtypes: float64(1), int64(4), str(13)
memory usage: 835.9 KB
```

```
[6]: df.describe()
```

[6]:	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
	count	3900.000000	3900.000000	3863.000000	3900.000000
	mean	1950.500000	44.068462	59.764359	3.750065
	std	1125.977353	15.207589	23.685392	0.716983
	min	1.000000	18.000000	20.000000	2.500000
	25%	975.750000	31.000000	39.000000	3.100000
	50%	1950.500000	44.000000	60.000000	3.800000
	75%	2925.250000	57.000000	81.000000	4.400000
	max	3900.000000	70.000000	100.000000	5.000000

- Finding first and last 5 values using `df.head(5)` and `df.tail(5)`.

```
[3]: df.head()
```

[3]:

Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases	
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Venmo	Fortnightly
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash	Fortnightly
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Credit Card	Weekly
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	PayPal	Weekly
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	PayPal	Annually

```
[4]: df.tail()
```

[4]:

Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases	
3895	3896	40	Female	Hoodie	Clothing	28	Virginia	L	Turquoise	Summer	4.2	No	2-Day Shipping	No	No	32	Venmo	Weekly
3896	3897	52	Female	Backpack	Accessories	49	Iowa	L	White	Spring	4.5	No	Store Pickup	No	No	41	Bank Transfer	Bi-Weekly
3897	3898	46	Female	Belt	Accessories	33	New Jersey	L	Green	Spring	2.9	No	Standard	No	No	24	Venmo	Quarterly
3898	3899	44	Female	Shoes	Footwear	77	Minnesota	S	Brown	Summer	3.8	No	Express	No	No	24	Venmo	Weekly
3899	3900	52	Female	Handbag	Accessories	81	California	M	Beige	Spring	3.1	No	Store Pickup	No	No	33	Venmo	Quarterly

- Finding null values that is present in the dataset using `df.isnull().sum()`.

Finding Missing Values

```
[7]: df.isnull().sum()
```

```
[7]: Customer ID          0
     Age                0
     Gender             0
     Item Purchased     0
     Category           0
     Purchase Amount (USD) 0
     Location           0
     Size               0
     Color              0
     Season             0
     Review Rating      37
     Subscription Status 0
     Shipping Type      0
     Discount Applied   0
     Promo Code Used    0
     Previous Purchases 0
     Payment Method     0
     Frequency of Purchases 0
     dtype: int64
```

- Replacing null values with the median of each category rating.

Replacing Null Value with median of a category

```
[8]: df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))
```

```
[9]: df.isnull().sum()
```

```
[9]: Customer ID          0
     Age                0
     Gender             0
     Item Purchased     0
     Category           0
     Purchase Amount (USD) 0
     Location           0
     Size               0
     Color              0
     Season             0
     Review Rating      0
     Subscription Status 0
     Shipping Type      0
     Discount Applied   0
     Promo Code Used    0
     Previous Purchases 0
     Payment Method     0
     Frequency of Purchases 0
     dtype: int64
```

- Change the columns name into snake case(in lower case and _ in place of ' ').

Modify columns in snake case

```
[9]: df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')
df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})

[10]: df.columns

[10]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
        'purchase_amount', 'location', 'size', 'color', 'season',
        'review_rating', 'subscription_status', 'shipping_type',
        'discount_applied', 'promo_code_used', 'previous_purchases',
        'payment_method', 'frequency_of_purchases'],
        dtype='str')

[11]: # create a column age_group
labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels=labels)

[12]: df[['age', 'age_group']].head(10)

[12]:
```

	age	age_group
0	55	Middle-aged
1	19	Young Adult
2	50	Middle-aged
3	21	Young Adult
4	45	Middle-aged
5	46	Middle-aged
6	63	Senior
7	27	Young Adult
8	26	Young Adult
9	57	Middle-aged

- In above we convert frequency into days.

```
[13]: # create purchase_freq_days columns
frequency_mapping = {
    'Fortnightly': 14,
    'Weekly': 7,
    'Monthly': 30,
    'Quarterly': 90,
    'Bi-Weekly': 14,
    'Annually': 365,
    'Every 3 Months': 90
}
df['purchase_freq_days'] = df['frequency_of_purchases'].map(frequency_mapping)

[14]: df[['purchase_freq_days', 'frequency_of_purchases']].head(10)

[14]:
```

	purchase_freq_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually
5	7	Weekly
6	90	Quarterly
7	7	Weekly
8	365	Annually
9	90	Quarterly

- Checking whether discount_applied and promo_code_used, if found duplicate then remove the

```
[15]: df[['discount_applied', 'promo_code_used']].head(10)
```

```
[15]:
```

	discount_applied	promo_code_used
0	Yes	Yes
1	Yes	Yes
2	Yes	Yes
3	Yes	Yes
4	Yes	Yes
5	Yes	Yes
6	Yes	Yes
7	Yes	Yes
8	Yes	Yes
9	Yes	Yes

```
[16]: # Checking redundancy of columns
```

```
(df['discount_applied']==df['promo_code_used']).all()
```

```
[16]: np.True_
```

- Drop column named promo_code_applied and installing psycpg2 and sqlalchemy.

```
[16]: # Checking redundancy of columns
```

```
(df['discount_applied']==df['promo_code_used']).all()
```

```
[16]: np.True_
```

```
[17]: # removing columns
```

```
df = df.drop('promo_code_used', axis=1)
```

```
[18]: df.columns
```

```
[18]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
        'purchase_amount', 'location', 'size', 'color', 'season',
        'review_rating', 'subscription_status', 'shipping_type',
        'discount_applied', 'previous_purchases', 'payment_method',
        'frequency_of_purchases', 'age_group', 'purchase_freq_days'],
        dtype='str')
```

```
[21]: pip install psycpg2-binary sqlalchemy
```

```
Requirement already satisfied: psycpg2-binary in C:\Users\Nazim Khan\AppData\Local\Programs\Python\Python311\Lib\site-packages (2.9.12)
```

```
Requirement already satisfied: sqlalchemy in C:\Users\Nazim Khan\AppData\Local\Programs\Python\Python311\Lib\site-packages (2.0.51)
```

```
Requirement already satisfied: greenlet>=1 in C:\Users\Nazim Khan\AppData\Local\Programs\Python\Python311\Lib\site-packages (from sqlalchemy) (3.5.1)
```

```
Requirement already satisfied: typing-extensions>=4.6.0 in C:\Users\Nazim Khan\AppData\Local\Programs\Python\Python311\Lib\site-packages (from sqlalchemy) (4.15.0)
```

```
Note: you may need to restart the kernel to use updated packages.
```

```
[notice] A new release of pip is available: 26.1.2 -> 26.2
```

```
[notice] To update, run: C:\Users\Nazim Khan\AppData\Local\Programs\Python\Python311\python.exe -m pip install --upgrade pip
```

- Connecting the processed data to PostgreSQL Database and save the data.

connecting PostgreSQL to Notebook

```
[26]: from sqlalchemy import create_engine

# connect to postgresql
username = "postgres"
password = "7290"
host = "localhost"
port = "5432"
database = "Customer_Behaviour"

engine = create_engine(f"postgresql+psycopg2://{username}:{password}@{host}:{port}/{database}")

# Step 2 Load Dataframe into PostgreSQL
table_name = "customer" # table name
df.to_sql(table_name, engine, if_exists="replace", index=False)

print(f"Data successfully loaded into table '{table_name}' in database '{database}',")

Data successfully loaded into table 'customer' in database 'Customer_Behaviour',
```

[]:

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

- 4.1. Revenue by Gender – Compared total revenue generated by male vs. female customers.

	gender 	revenue 
1	Female	75191
2	Male	157890

- 4.2. High-Spending Discount Users – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id bigint 	purchase_amount bigint 
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94
13	32	79
14	33	67
15	35	91
16	37	69
17	40	60
18	41	76

4.3. Top 5 Products by Rating – Found products with the highest average review ratings.

	item_purchased text 	Average Product Rating numeric 
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

- 4.4. Shipping Type Comparison – Compared average purchase amounts between Standard and Express shipping.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

- 4.5. Subscribers vs. Non-Subscribers – Compared average spend and total revenue across subscription status.

	subscription_status text	Total Customers bigint	Average Spend numeric	Total Revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

- 4.6. Discount-Dependent Products – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased text	Discount Rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

- 4.7. Customer Segmentation – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

- 4.8. Top 3 Products per Category – Listed the most purchased products within each category.

	item_rank bigint	category text	item_purchased text	Total Orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

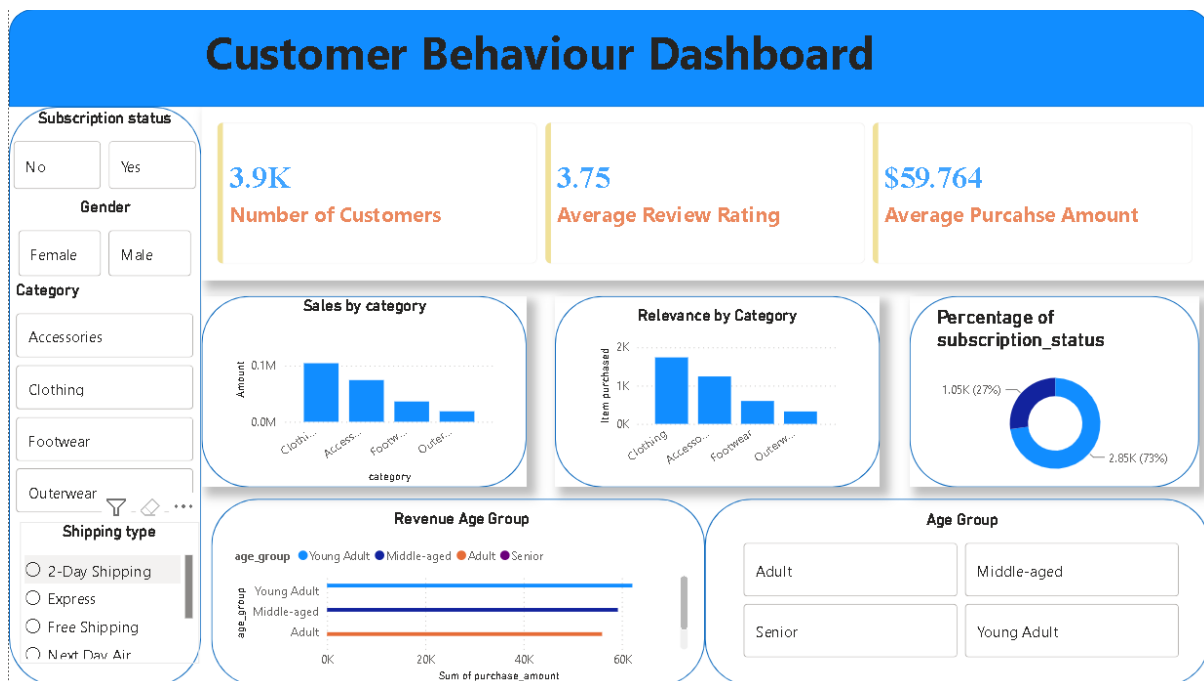
4.9. Repeat Buyers & Subscriptions – Checked whether customers with >5 purchases are more likely to subscribe.

	subscription_status text	Repeat buyers bigint
1	No	2518
2	Yes	958

4.10. Revenue by Age Group – Calculated total revenue contribution of each age group. 5.

	age_group text	Total Revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Dashboard in Power BI Finally, we built an interactive dashboard in Power BI to present insights visually.



6. Business Recommendations

- Boost Subscriptions – Promote exclusive benefits for subscribers.
- Customer Loyalty Programs – Reward repeat buyers to move them into the “Loyal” segment.
- Review Discount Policy – Balance sales boosts with margin control.
- Product Positioning – Highlight top-rated and best-selling products in campaigns.
- Targeted Marketing – Focus efforts on high-revenue age groups and express-shipping users.